MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

BUFFER-LIMITER, ANALOG, KEY CODE BAC

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic buffer-limiter module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

* 3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.35 failures per million operating hours when predicted in accordance with MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be BAC.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 30 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Pin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/190-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

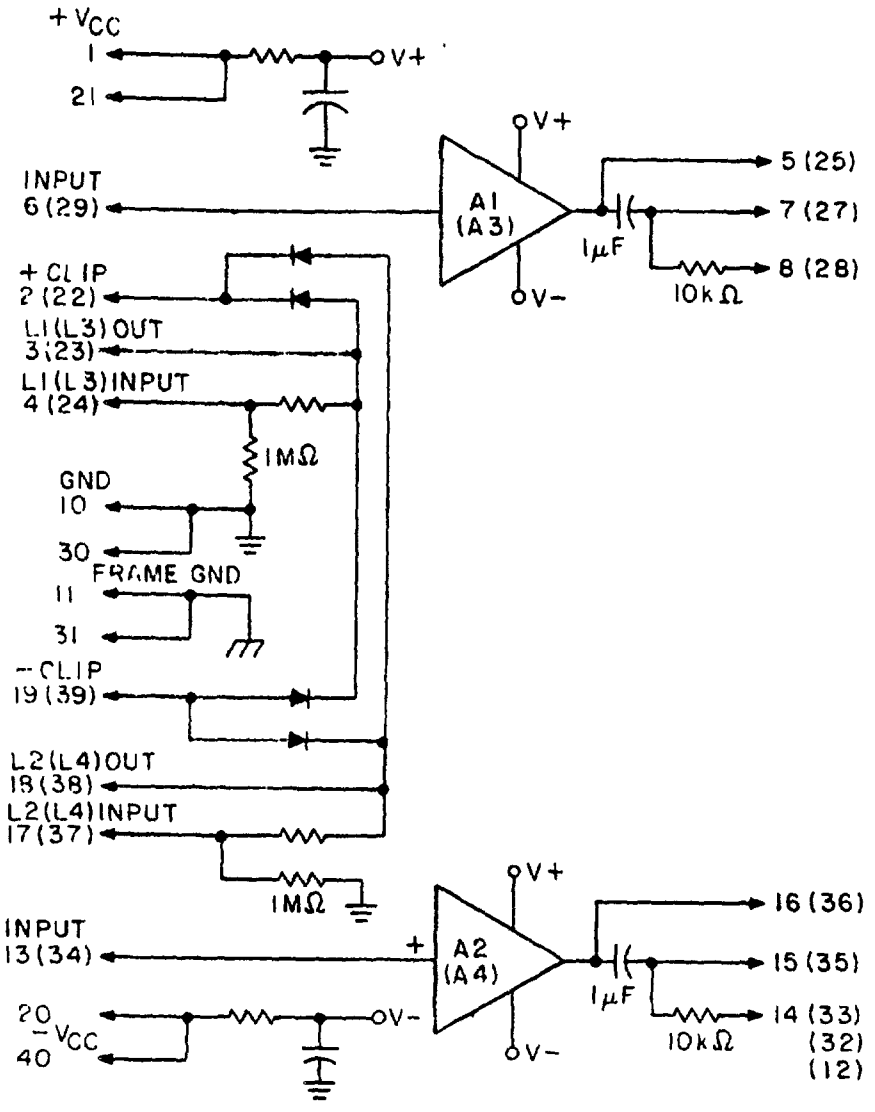
4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirement of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through AA shall be performed in any order with the following exceptions. If test conditions A, B, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

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NOTES:
1. Only one-half of the module diagram is shown. The other half is represented by the pin numbers in parentheses.
2. Unused pins: 9 and 26.
3. A1-A4 = Amplifiers.
4. L1-L4 = Limiters.

FIGURE 1. BAC module function diagram, buffer-limiter.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1, A2	Isolation	II	S	100	∞	10	∞	100	∞	10	∞	MΩ
B	Stress	III	Q	1	2	1	2	1	2	1	2	Minute
C1-C7	Continuity	IV	S	0	1.0	0	1.0	0	1.0	0	1.0	Ω
D	Output short circuit	V	Q	10	15	10	15	10	15	10	15	Minute
E1-E4	Offset voltage	VI	X	-5.0	5.0	-6.0	6.0	-7.0	7.0	-8.0	8.0	mVdc
F1-F4	Offset voltage drift with temperature check	VI	Q					0	20	0	30	µV/°C
G1-G4	Bias current (I_b) check	VI	X	0	4.0	0	6.0	0	12	0	20	nAdc
* H1-H4	Limiter input resistance	VII	S	0.85	1.10	0.85	1.10	0.80	1.10	0.55	1.20	MΩ
J1-J4	Output resistance	VII	S	9.5	10.5	9.0	11.0	9.2	10.8	8.0	12	kΩ
K1-K4	Output capacitance	VII	S	0.9	1.1	0.8	1.2	0.8	1.2	0.5	1.5	µF
L1, L11, L21, L31	Limiter input current check	VIII	Q	-9.3	-7.6	-9.3	-7.6	-9.5	-7.4	-9.5	-7.4	mAdc
L2, L12, L22, L32	Limiter input current check	VIII	Q	-4.5	-3.3	-4.5	-3.3	-4.6	-3.2	-4.6	-3.2	mAdc
L3, L13, L23, L33	Limiter input current check	VIII	Q	-0.50	0.00	-0.50	0.00	-0.50	0.00	-0.50	0.00	mAdc
L4, L14, L24, L34	Limiter input current check	VIII	Q	-0.010	0.00	-0.010	0.00	-0.010	0.00	-0.012	0.00	mAdc
L5, L15, L25, L35	Limiter input current check	VIII	Q	-0.005	0.00	-0.005	0.00	-0.005	0.00	-0.006	0.00	mAdc
L6, L16, L26, L36	Limiter input current check	VIII	Q	0.00	0.005	0.00	0.005	0.00	0.005	0.00	0.006	mAdc
L7, L17, L27, L37	Limiter input current check	VIII	Q	0.00	0.010	0.00	0.010	0.00	0.010	0.00	0.012	mAdc
L8, L18, L28, L38	Limiter input current check	VIII	Q	0.00	0.50	0.00	0.50	0.00	0.50	0.00	0.50	mAdc

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
L9, L19, L29, L39	Limiter input current check	VIII	Q	3.3	4.5	3.3	4.5	3.2	4.6	3.2	4.6	mAdc
L10, L20, L30, L40	Limiter input current check	VIII	Q	7.6	9.3	7.6	9.3	7.4	9.5	7.4	9.5	mAdc
M1, M9, M17, M25	Limiter output voltage	VIII	Q	-6.10	-4.95	-6.10	-4.95	-6.10	-4.95	-6.10	-4.95	Vdc
M2, M10, M18, M26	Limiter output voltage	VIII	Q	-5.55	-4.95	-5.60	-4.95	-5.60	-4.95	-5.65	-4.95	Vdc
M3, M11, M19, M27	Limiter output voltage	VIII	Q	-5.05	-4.95	-5.05	-4.95	-5.05	-4.95	-5.05	-4.95	Vdc
M4, M12, M20, M28	Limiter output voltage	VIII	Q	-2.01	-1.99	-2.01	-1.99	-2.01	-1.99	-2.01	-1.99	Vdc
M5, M13, M21, M29	Limiter output voltage	VIII	Q	1.99	2.01	1.99	2.01	1.99	2.01	1.99	2.01	Vdc
M6, M14, M22, M30	Limiter output voltage	VIII	Q	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	Vdc
M7, M15, M23, M31	Limiter output voltage	VIII	Q	4.95	5.55	4.95	5.60	4.95	5.60	4.95	5.65	Vdc
M8, M16, M24, M32	Limiter output voltage	VIII	Q	4.95	6.10	4.95	6.10	4.95	6.10	4.95	6.10	Vdc
N1, N13, N25, N37	Limiter out for positive clip	IX	Q	-15.0	-13.9	-15.0	-13.9	-15.0	-13.9	-15.0	-13.9	Vdc
N2, N14, N26, N38	Limiter out for positive clip	IX	Q	-0.50	0.00	-0.50	0.00	-0.50	0.00	-0.50	0.00	Vdc
N3-N6, N15-N18, N27-N30, N39-N42	Limiter out for positive clip	IX	Q	-0.005	+0.005	-0.005	+0.005	-0.005	+0.005	-0.005	+0.005	Vdc
N7-N10, N19-N22, N31-N34, N43-N46	Limiter out for negative clip	IX	Q	-0.005	+0.005	-0.005	+0.005	-0.005	+0.005	-0.005	+0.005	Vdc

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N11, N23, N35, N47	Limiter out for negative clip	IX	Q	0.00	0.50	0.00	0.50	0.00	0.50	0.00	0.50	Vdc
N12, N24, N36, N48	Limiter out for negative clip	IX	Q	13.9	15.0	13.9	15.0	13.9	15.0	13.9	15.0	Vdc
P1, P2	Power supply current	X	Q	5	24	5	28	5	32	5	36	mAdc
P3, P4	Power supply current	X	Q	5	26	5	30	5	34	5	38	mAdc
Q1-Q16	Small signal voltage gain check	XI	S	0.95	1.05	0.95	1.05	0.95	1.05	0.95	1.05	V/V
R1-R4	Large signal voltage gain check	XII	X	0.95	1.05	0.95	1.05	0.95	1.05	0.95	1.05	V/V
S1-S4	Output voltage swing	XII	X	7.0	15	7.0	15	7.0	15	7.0	15	Vpk
V1-V4	Input resistance	XIII	S	5	∞	5	∞	5	∞	5	∞	MΩ
W1-W4	Output noise	XIV	S	0.0	50	0.0	50	0.0	50	0.0	50	μVrms
X1-X8	Power supply rejection ratio check	XV	Q	60	∞	60	∞	60	∞	60	∞	dB
Y1-Y4	Functional without input limiting	XVI	X	2.40	2.60	2.40	2.60	2.40	2.60	2.40	2.60	Vrms
Y5-Y8	Functional with input limiting	XVI	X	4.9	6.3	4.9	6.3	4.9	6.3	4.9	6.3	Vpk
Z1-Z12	Crosstalk	XVII	Q	60	∞	60	∞	60	∞	60	∞	dB
AA1, AA2	Power supply transient rise time	XVIII	S	31	93	31	93	31	93	31	93	μs

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through AA in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E through AA shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests E through AA shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests E through AA shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Part number M28787/190-1 supersedes part number M28787/190. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Supply voltage	+15 Vdc ±10 percent -15 Vdc ±10 percent
Supply current Quiescent Maximum signal, maximum load, undistorted output (e_{in} = 7 Vrms, R_L = 2 kΩ)	 36 mAdc each supply 38 mAdc each supply
Gain	Unity (1) noninverting
Bandwidth, -3 dB	Greater than 100 kHz
Output. Voltage swing (R_L = 10 kΩ) Noise	 +7 V_{pk} 50 μVrms maximum

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Input impedance (buffer amplifiers)	5 MΩ minimum
Power supply rejection ratio	60 dB
Power dissipation	
Quiescent	1.2 W
Maximum signal, maximum load	1.3 W

6.5 Module function. This module contains four voltage-follower amplifiers with optional pin programmable input signal limiting.

6.5.1 Amplifiers. The four amplifiers are electrically independent of the optional clipping circuitry. To use the limiting capability of the module, appropriate external connections are made. Thus, each amplifier is used as an independent voltage follower. Each amplifier is an internally compensated unity-gain noninverting amplifier. No external compensation is required.

6.5.2 Input limiting. To use the limiting circuitry, proper connections are made externally.

Amplifier	Connect	Limited input
No. 1	Pin 3 to pin 6	Pin 4
No. 2	Pin 13 to pin 18	Pin 17
No. 3	Pin 23 to pin 29	Pin 24
No. 4	Pin 34 to pin 38	Pin 37

6.5.2.1 Clipping circuitry. The clipping circuitry for each amplifier presents a load of 1 MΩ to the input signal when not clipping. When clipping, the load is approximately 1 kΩ.

6.5.2.2 Limiting circuitry. The limiting circuits for amplifier No. 1 and amplifier No. 2 use the same clipping reference voltages. Likewise, the limiters for amplifier No. 3 and amplifier No. 4 use the same voltages (see figure 1). The plus and minus clipping voltages do not have to be equal. They must be equal to or less than 10 V in magnitude. The input signal is clipped at a voltage whose magnitude is approximately equal to the applied reference voltage, plus the voltage drop across a silicon diode.

6.5.3 Output options. Each amplifier has a direct output and a capacitively coupled output (see figure 1). A 10 kΩ resistor is provided for known output loading, if desired.

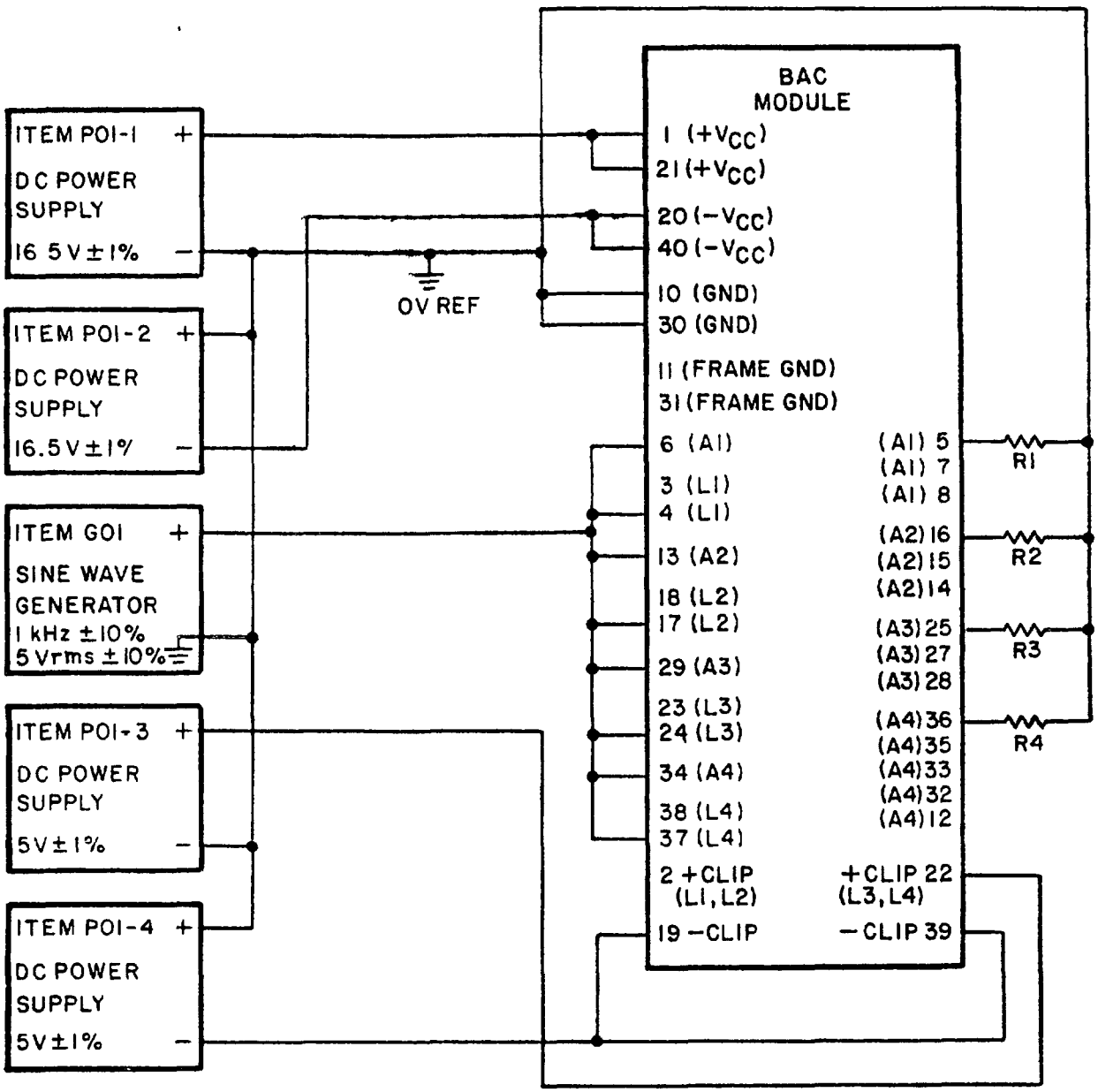
6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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NOTE: R1-R4 = 10 kΩ ±1%, 1/8 W.

FIGURE 2. Operating temperature and life test circuit.

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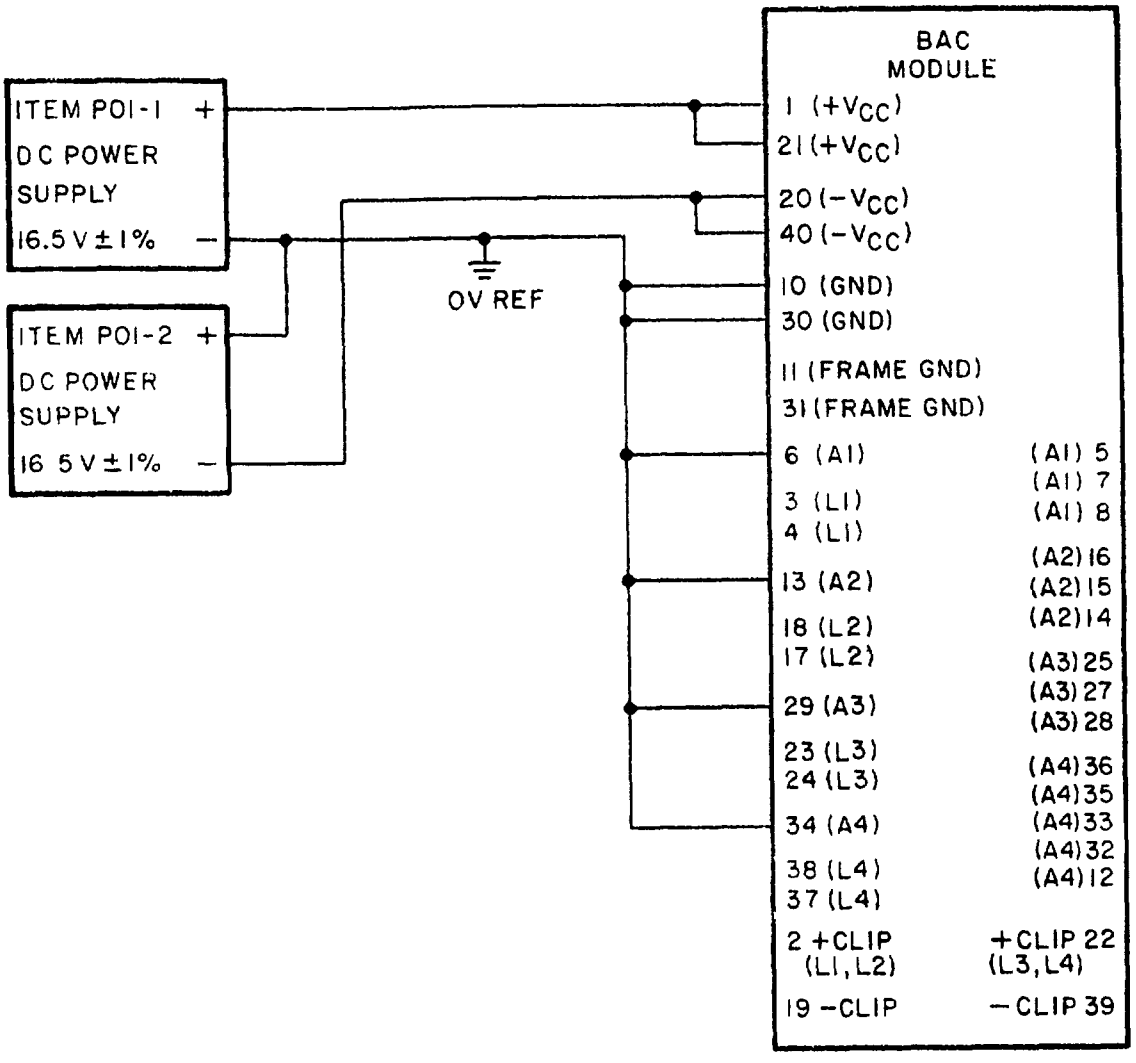


FIGURE 3. Component temperature test circuit

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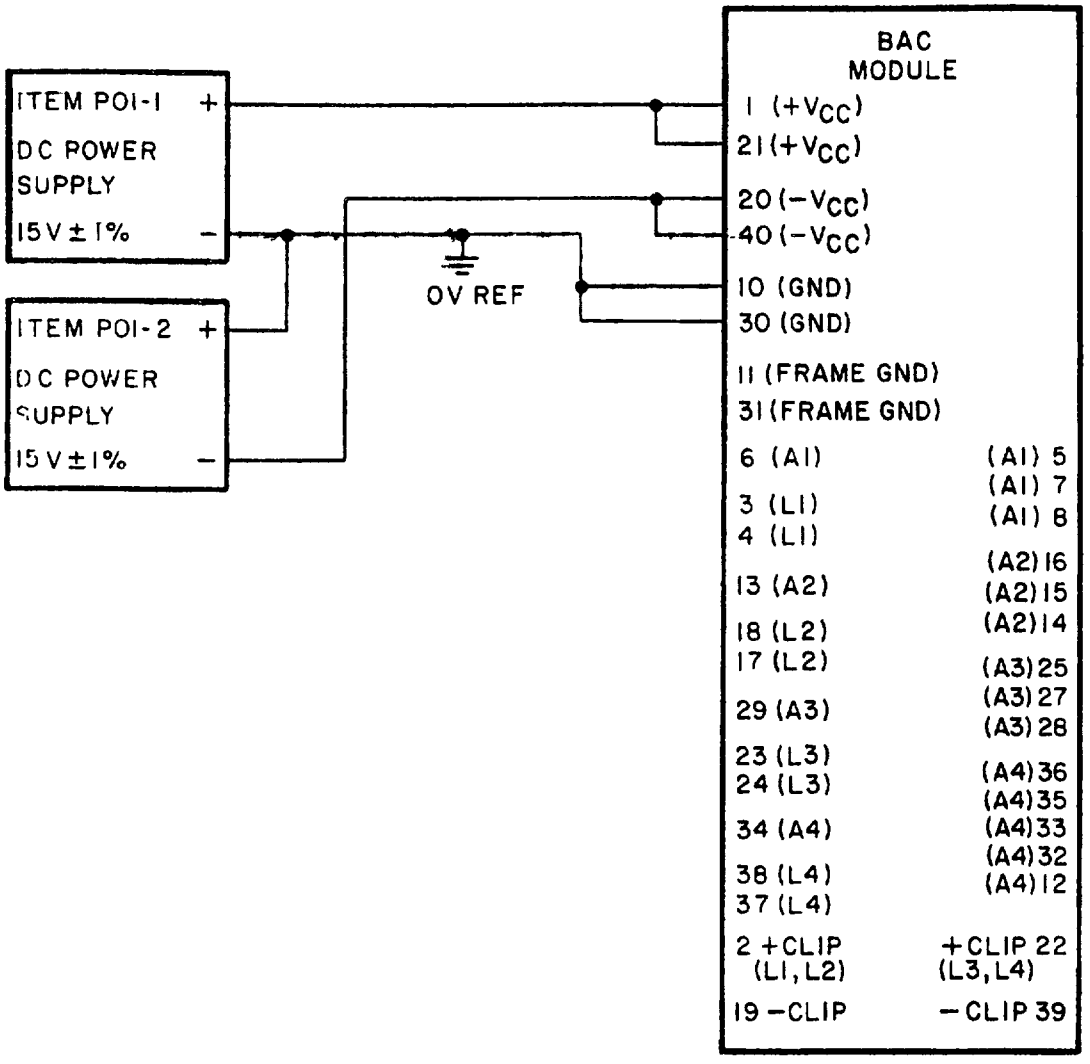


FIGURE 4. Vibration test circuit

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TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	9, 26	All other pins	100	∞	MΩ	5
A2	Isolation	11, 31	All other pins	100	∞	MΩ	5

TABLE III. Stress test.

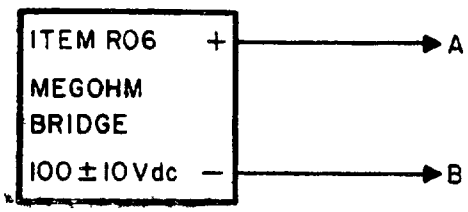
Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
B	Stress 1/	1	2	Minute	6

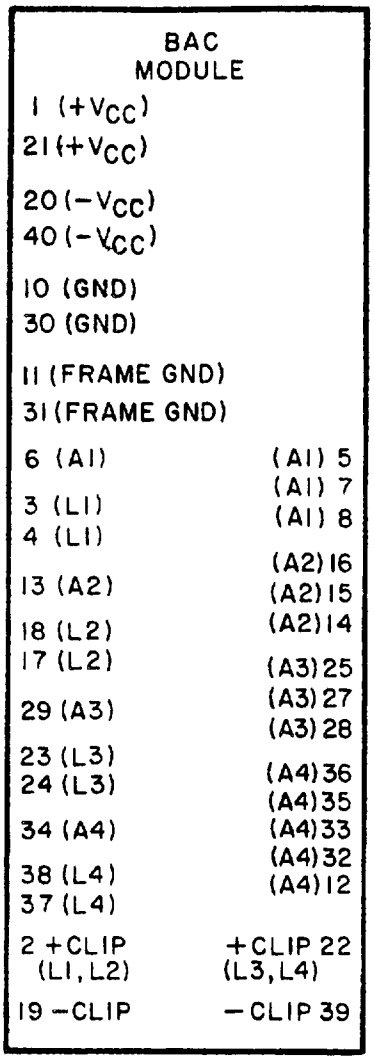
1/ Module shall conform to all electrical requirements after this test.

TABLE IV. Continuity test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Continuity	11	Frame	0	1.0	Ω	7
C2	↓	11	31	↓	↓	↓	↓
C3	↓	1	21	↓	↓	↓	↓
C4	↓	10	30	↓	↓	↓	↓
C5	↓	20	40	↓	↓	↓	↓
C6	↓	12	32	↓	↓	↓	↓
C7	↓	32	33	↓	↓	↓	↓

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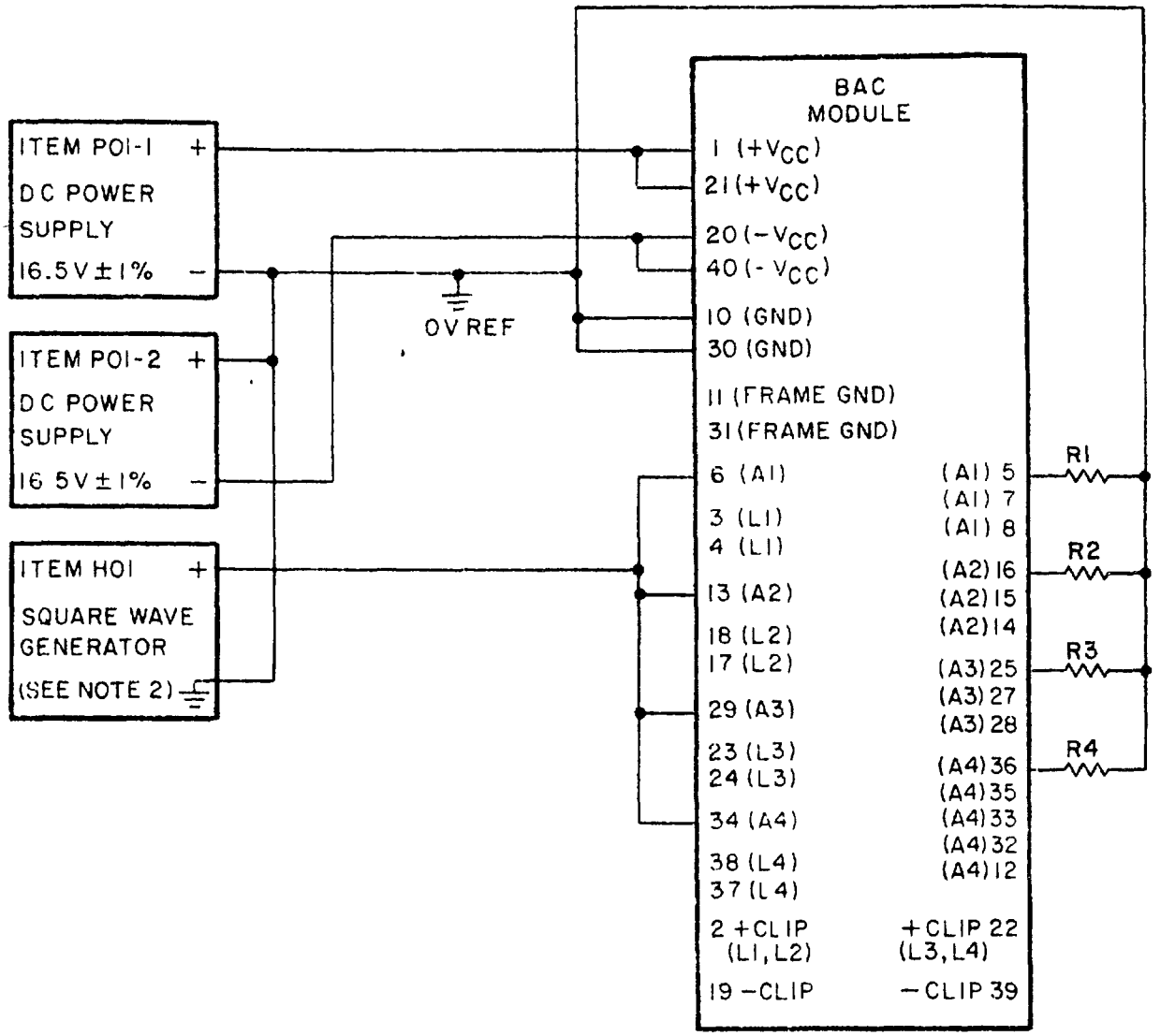


WARNING:

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

FIGURE 5. Isolation test circuit for table II.

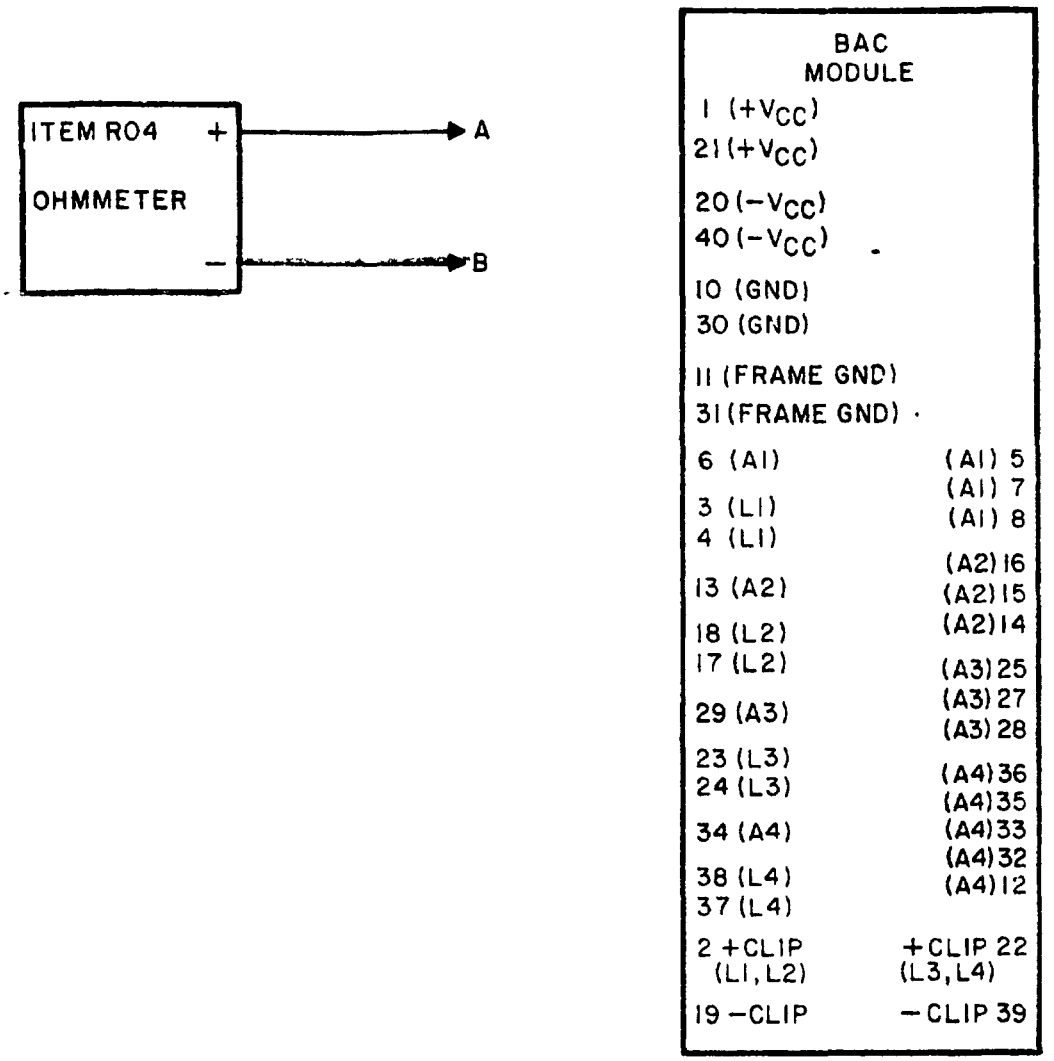
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NOTES:
1. R1-R4 = 10 kΩ ±1%, 1/8 W.
2. Item H01 output pulse parameters:
   PRR = 100 Hz ±10%.
   PA = 25 V_{p-p}.
   $t_{TLH} = t_{THL}$ = 1 µs maximum.
   Duty cycle = 50%.

FIGURE 6. Stress test circuit for table III.

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NOTE: For test condition C1 penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of item R04.

FIGURE 7. Continuity test circuit for table IV.

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TABLE V. Output short circuit test.

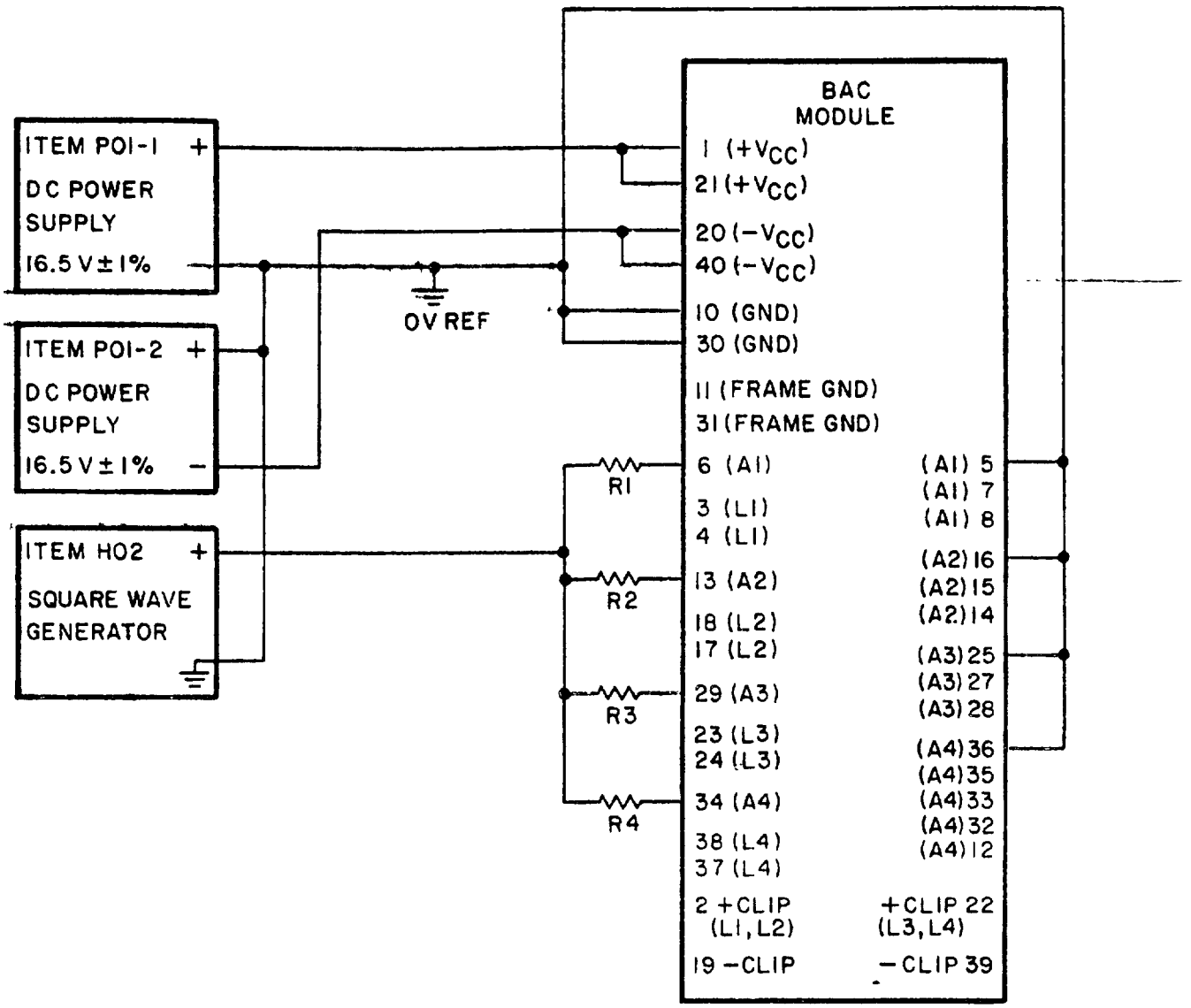
Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
D	Output short circuit 1/	10	15	Minute	8

1/ The module shall conform to the electrical requirements of test conditions C and E through AA after this test.

TABLE VI Offset voltage test, offset voltage drift with temperature check, and bias current (I_b) check.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	Offset voltage	6	5	-5.0	5.0	mVdc	9
E2		13	16				
E3		29	25				
E4		34	36				
F1	Offset voltage drift with temperature check	6	5	0	20	µV/°C	
F2		13	16				
F3		29	25				
F4		34	36				
G1	Bias current (I_b) check	6	5		4.0	nAdc	
G2		13	16				
G3		29	25				
G4		34	36				

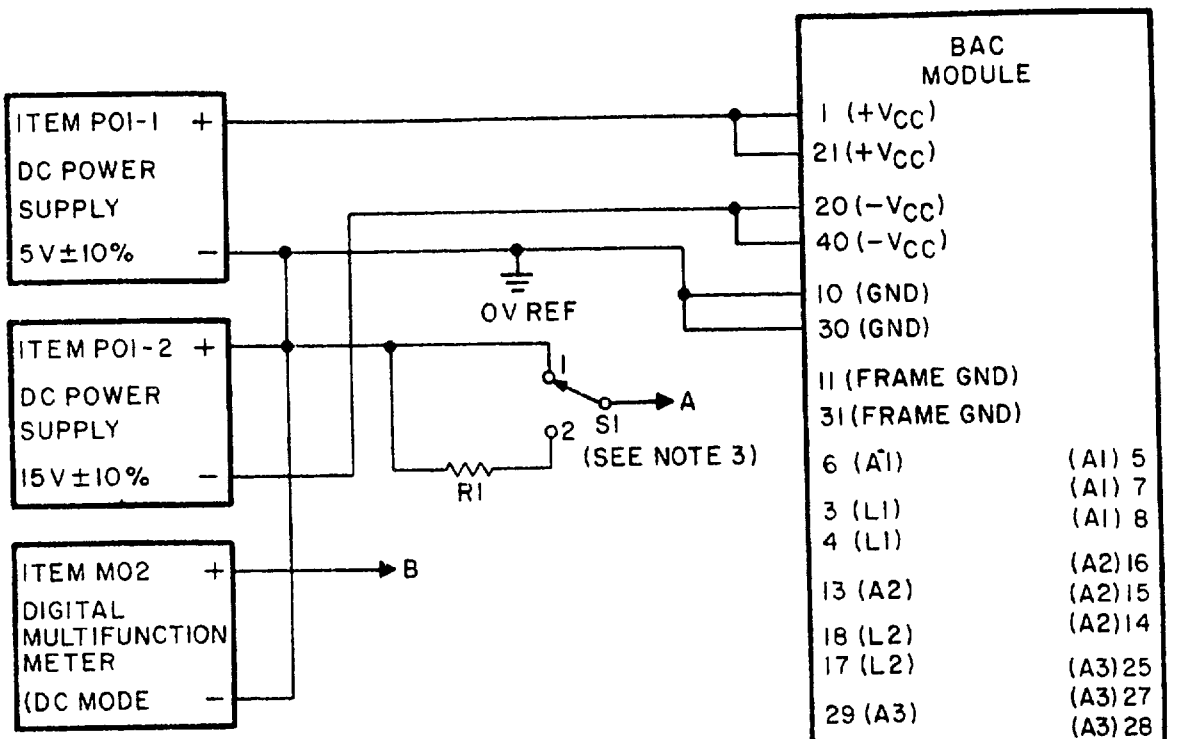
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NOTES:
1. R1-R4 = 5 kΩ ±10%, 1/2 W.
2. Item HO1 output pulse parameters:
   PRR = 100 Hz ±10%.
   PA = 25 V_{p-p}.
   $t_{TLH} = t_{THL}$ = 1 µs maximum.
   Duty cycle = 50%.

FIGURE 8. Output short circuit test for table V.

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NOTES
1. R1 = 1 MΩ ±1%, 1/8 W
2. S1 = SPDT switch.
3. Set switch S1 to position 1 for test conditions E1-E4 and F1-F4 and to position 2 for test conditions G1-G4.
4. V_{OUT} = -5.0 mV minimum, 5.0 mV maximum for test conditions E1-E4
5. ΔV_{OUT} = -4.0 mV minimum, 4.0 mV maximum for test conditions G1-G4.
6. Drift = $\frac{[V_{OS}\ (t = 0^{o}C) - V_{OS}\ (t = 60^{o}C)]}{60^{o}C}$
7. $I_b = \frac{[V_{OUT}\ (S1 = \text{Position 1}) - V_{OUT}\ (S1 = \text{Position 2})]}{1\ M\Omega}$

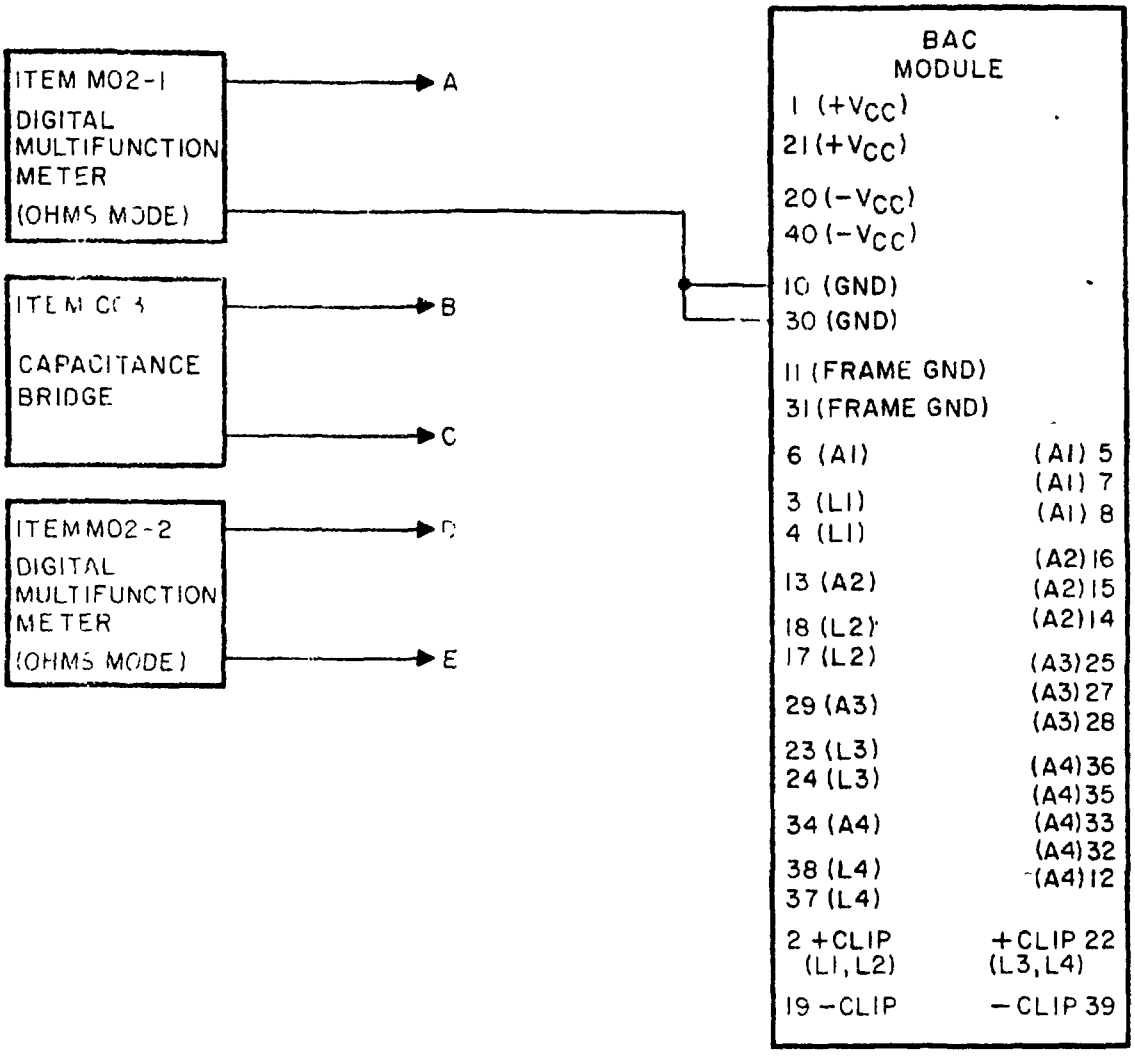
*FIGURE 9. Offset voltage test, offset voltage drift with temperature check, and bias current (I_b) check test circuit for table VI.

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TABLE VII. Limiter input resistance, output resistance, and output capacitance test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
H1	Limiter input resistance	4	Open	Open	Open	Open	0.85	1.10	MΩ	10
H2		17								
H3		24								
H4		37								
J1	Output resistance	Open			7	8	9.5	10.5	kΩ	
J2					14	15				
* J3					27	28				
J4					33	35				
K1	Output capacitance		5	7	Open	Open	0.9	1.1	µF	
K2			15	16						
K3			25	27						
K4			35	36						

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NOTES
1. Item C03 setup.
   f = 1 kHz ±10%.
   V_{pk} = 1 V maximum.
2. Since items M02-1 and M02-2 are not used simultaneously, only one is required.
3. For test conditions H1-H4, item M02 shall be set to a range which measures resistance with 1 µA current.

* FIGURE 10 Limiter input resistance, output resistance, and output capacitance test circuit for table VII.

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TABLE VIII. Limiter input current check and limiter output voltage test.

Test condition	Parameter		Item P01-1 V ±0.1%	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
				A to pin	B to pin	C to pin	D to pin	Min	Max		
L1	L1	Limiter input current check	-15.0	GND	E	E	4	-9.3	-7.6	mAdc	11
L2	L1		-10.0					-4.5	-3.3		
L3	L1		-5.50					-0.50	0.00		
L4	L1		-5.00					-0.010	0.00		
L5	L1		-2.00					-0.005	0.00		
L6	L1		+2.00	E	GND			0.00	0.005		
L7	L1		+5.00					0.00	0.010		
L8	L1		+5.50					0.00	0.50		
L9	L1		+10.0					3.3	4.5		
L10	L1		+15.0					7.6	9.3		
L11	L2		-15.0	GND	F	F	17	-9.3	-7.6		
L12	L2		-10.0					-4.5	-3.3		
L13	L2		-5.50					-0.50	0.00		
L14	L2		-5.00					-0.010	0.00		
L15	L2		-2.00					-0.005	0.00		
L16	L2		+2.00	F	GND			0.00	0.005		
L17	L2		+5.00					0.00	0.010		
L18	L2		+5.50					0.00	0.50		
L19	L2		+10.0					3.3	4.5		
L20	L2		+15.0					7.6	9.3		

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TABLE VIII. Limiter input current check and limiter output voltage test - Continued.

Test condition	Parameter	Item P01-1 V ±0.1%	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
L21	L3 Limiter input current check	-15.0	GND	G	G	24	-9.3	-7.6	mAdc	11
L22	L3	-10.0					-4.5	-3.3		
L23	L3	-5.50					-0.50	0.00		
L24	L3	-5.00					-0.010	0.00		
L25	L3	-2.00					-0.005	0.00		
L26	L3	+2.00	G	GND			0.00	0.005		
L27	L3	+5.00					0.00	0.010		
L28	L3	+5.50					0.00	0.50		
L29	L3	+10.0					3.3	4.5		
L30	L3	+15.0					7.6	9.3		
L31	L4	-15.0	GND	H	H	37	-9.3	-7.6		
L32	L4	-10.0					-4.5	-3.3		
L33	L4	-5.50					-0.50	0.00		
L34	L4	-5.00					-0.010	0.00		
L35	L4	-2.00					-0.005	0.00		
L36	L4	+2.00	H	GND			0.00	0.005		
L37	L4	+5.00					0.00	0.010		
L38	L4	+5.50					0.00	0.50		
L39	L4	+10.0					3.3	4.5		
L40	L4	+15.0					7.6	9.3		

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TABLE VIII. Limiter input current check and limiter output voltage test - Continued.

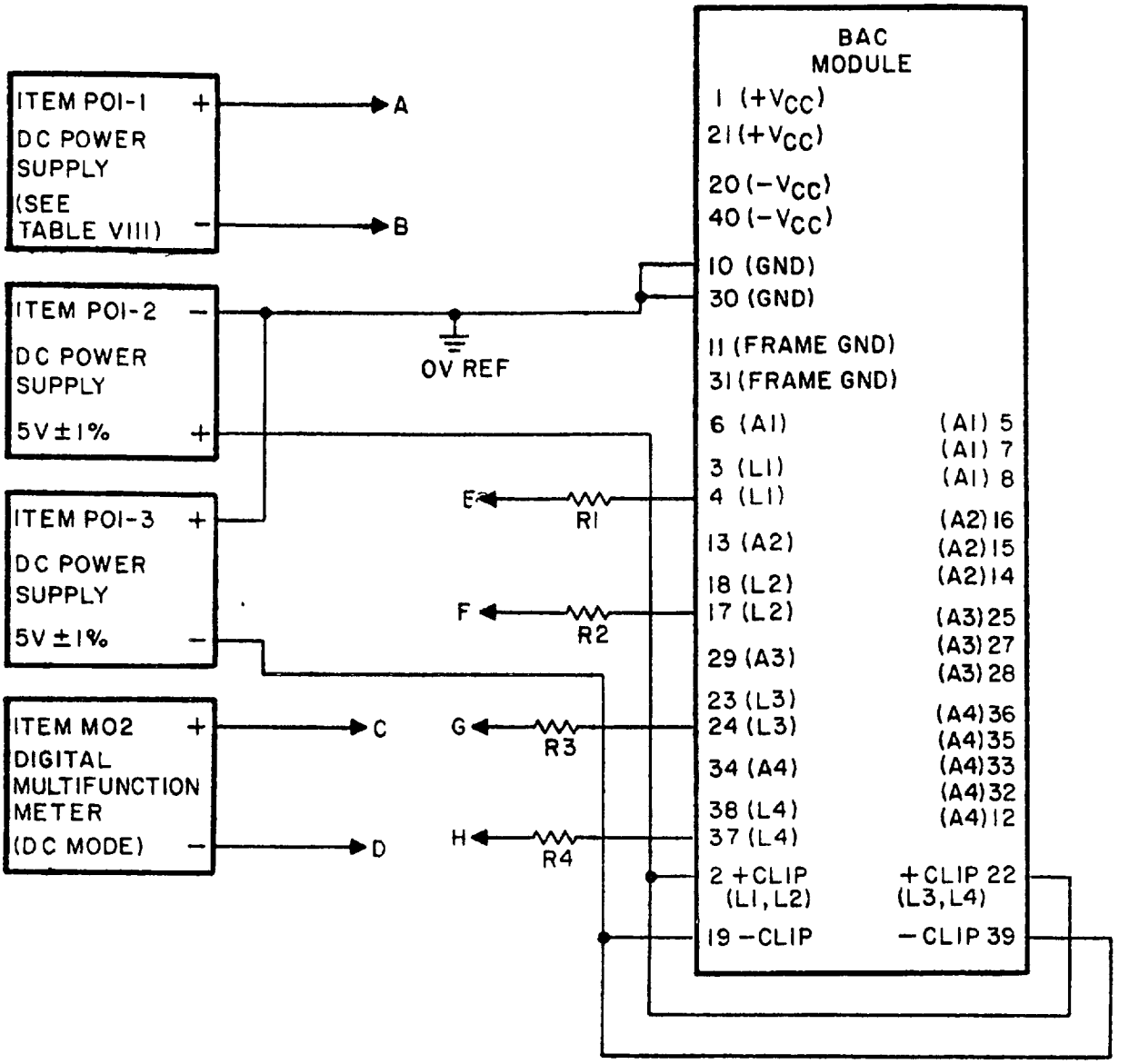
Test condition	Parameter	Item PO1-1 V ±0.1%	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
M1	L1 Limiter output voltage	-15.0	GND	E	3	GND	-6.10	-4.95	Vdc	11
M2	L1	-5.50					-5.55	-4.95		
M3	L1	-5.00					-5.05	-4.95		
M4	L1	-2.00					-2.01	-1.99		
M5	L1	2.00	E	GND			1.99	2.01		
M6	L1	5.00					4.95	5.05		
M7	L1	5.50					4.95	5.55		
M8	L1	15.0					4.95	6.10		
M9	L2	-15.0	GND	F	18		-6.10	-4.95		
M10	L2	-5.50					-5.55	-4.95		
M11	L2	-5.00					-5.05	-4.95		
M12	L2	-2.00					-2.01	-1.99		
M13	L2	2.00	F	GND			1.99	2.01		
M14	L2	5.00					4.95	5.05		
M15	L2	5.50					4.95	5.55		
M16	L2	15.0					4.95	6.10		
M17	L3	-15.0	GND	G	23		-6.10	-4.95		
M18	L3	-5.50					-5.55	-4.95		
M19	L3	-5.00					-5.05	-4.95		
M20	L3	-2.00					-2.01	-1.99		

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TABLE VIII. Limiter input current check and limiter output voltage test - Continued.

Test condition	Parameter	Item P01-1 V ±0.1%	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
M21	L3 Limiter output voltage	2.00	G	GND	23	GND	1.99	2.01	Vdc	11
M22	L3	5.00					4.95	5.05		
M23	L3	5.50					4.95	5.55		
M24	L3	15.0					4.95	6.10		
M25	L4	-15.0	GND	H	38		-6.10	-4.95		
M26	L4	-5.50					-5.55	-4.95		
M27	L4	-5.00					-5.05	-4.95		
M28	L4	-2.00					-2.01	-1.99		
M29	L4	2.00	H	GND			1.99	2.01		
M30	L4	5.00					4.95	5.05		
M31	L4	5.50					4.95	5.55		
M32	L4	15.0					4.95	6.10		

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NOTES:
1. R1-R4 = 100Ω ±1%, 1/8 W.
2. Limiter input current = $\frac{V(R = 100\Omega)}{100\Omega}$

*FIGURE 11. Limiter input current check and limiter output voltage test circuit for table VIII.

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TABLE IX. Limiter out for positive and negative clip test.

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Item P01 V ±0.1%	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
N1	L1	Limiter out for positive clip	GND	2	3	-15.0	-15.0	-13.9	Vdc	12
N2	L1		GND	2		-0.50	-0.50	0.00		
N3	L1		2	GND		0.00	-0.005	+0.005		
N4	L1					0.50				
N5	L1					5.00				
N6	L1					15.0				
N7	L1	Limiter out for negative clip	GND	19		-15.0				
N8	L1		GND	19		-5.00				
N9	L1		GND	19		-0.50				
N10	L1		19	GND		0.00				
N11	L1		19	GND		0.50	0.00	0.50		
N12	L1		19	GND		15.0	13.9	15.0		
N13	L2	Limiter out for positive clip	GND	2	18	-15.0	-15.0	-13.9		
N14	L2		GND	2		-0.50	-0.50	0.00		
N15	L2		2	GND		0.00	-0.005	+0.005		
N16	L2					0.50				
N17	L2					5.00				
N18	L2					15.0				

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TABLE IX. Limiter out for positive and negative clip test - Continued.

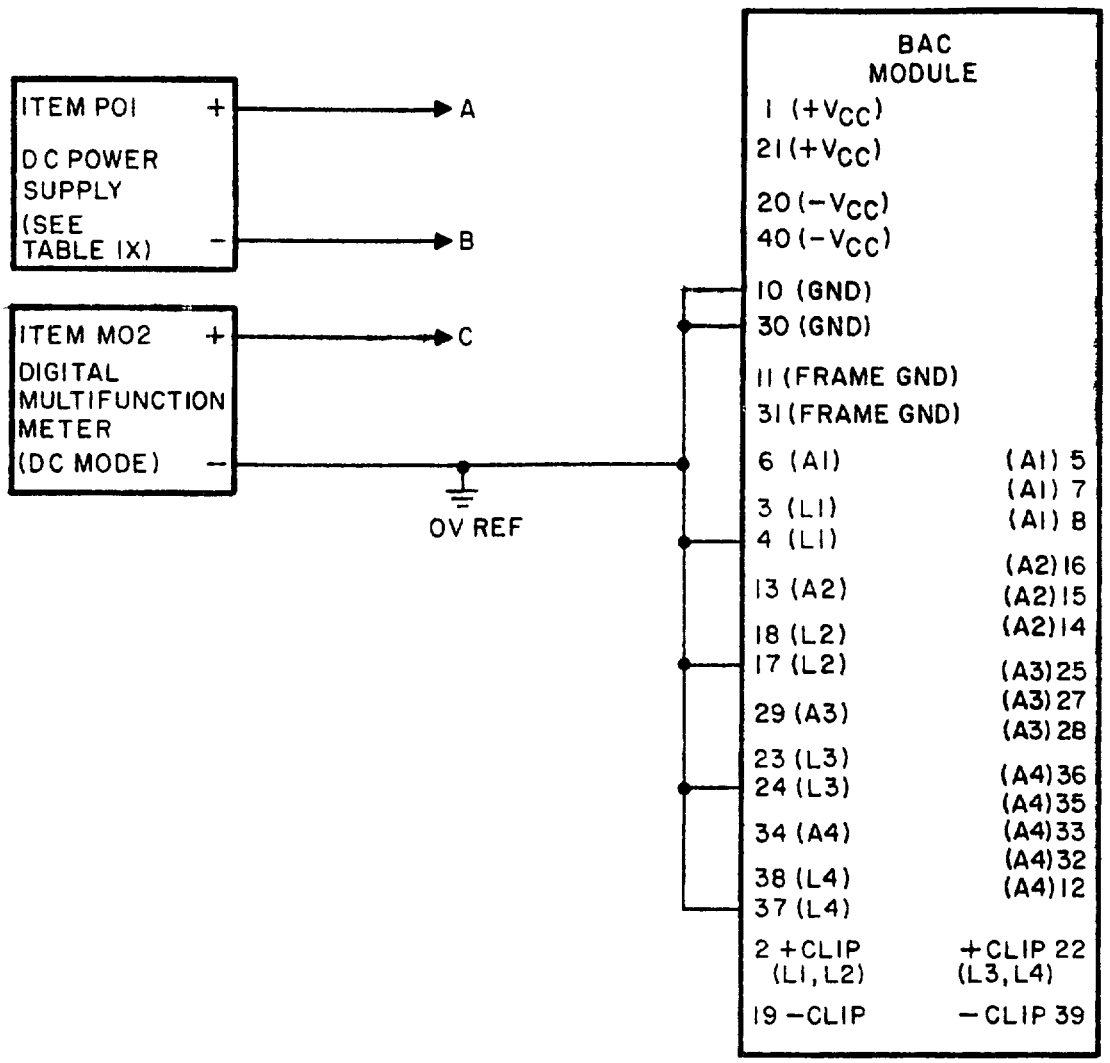
Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Item P01 V ±0.1%	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
N19	L2	Limiter out for negative clip	GND	19	18	-15.0	-0.005	+0.005	Vdc	12
N20	L2		GND	19		-5.00				
N21	L2		GND	19		-0.50				
N22	L2		19	GND		0.00				
N23	L2		19	GND		0.50	0.00	0.50		
N24	L2		19	GND		15.0	13.9	15.0		
N25	L3	Limiter out for positive clip	GND	22	23	-15.0	-15.0	-13.9		
N26	L3		GND	22		-0.50	-0.50	0.00		
N27	L3		22	GND		0.00	-0.005	+0.005		
N28	L3					0.50				
N29	L3					5.00				
N30	L3					15.0				
N31	L3	Limiter out for negative clip	GND	39		-15.0				
N32	L3		GND	39		-5.00				
N33	L3		GND	39		-0.50				
N34	L3		39	GND		0.00				
N35	L3		39	GND		0.50	0.00	0.50		
N36	L3		39	GND		15.00	13.9	15.0		

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TABLE IX. Limiter out for positive and negative clip test - Continued.

Test condition	Parameter		Connect			Item PO1 V ±0.1%	Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin		Min	Max		
N37	L4	Limiter out for positive clip	GND	22	38	-15.0	-15.0	-13.9	Vdc	12
N38	L4		GND	22		-0.50	-0.50	0.00		
N39	L4		22	GND		0.00	-0.005	+0.005		
N40	L4					0.50				
N41	L4					5.00				
N42	L4					15.0				
N43	L4	Limiter out for negative clip	GND	39		-15.0				
N44	L4		GND	39		-5.00				
N45	L4		GND	39		-0.50				
N46	L4		39	GND		0.00				
N47	L4		39	GND		0.50	0.00	0.50		
N48	L4		39	GND		15.0	13.9	15.0		

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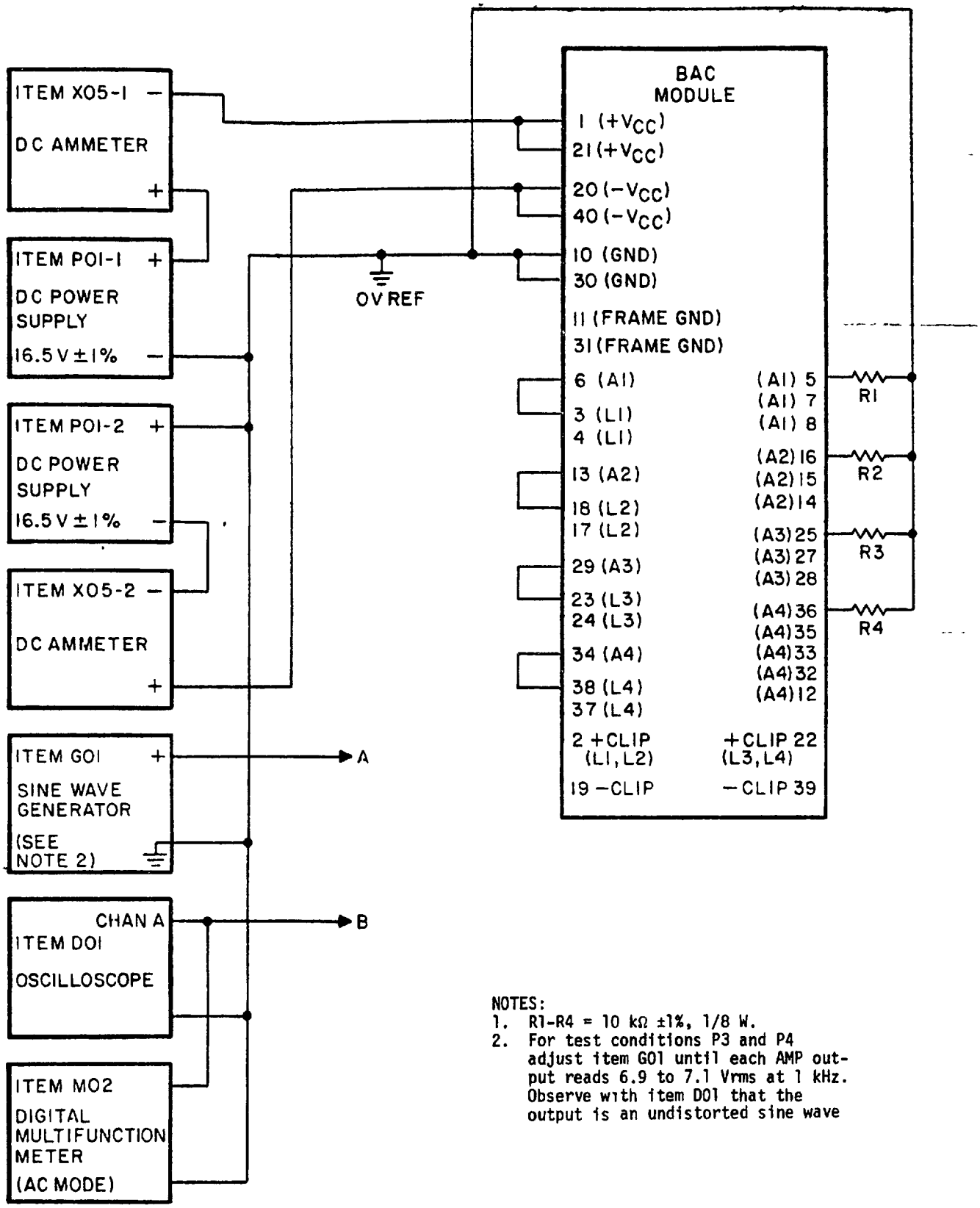
* FIGURE 12. Limiter out for positive and negative clip test circuit for table IX.

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TABLE X. Power supply current test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
P1	Power supply current	Open	Open	5	24	mAdc	13
P2	↓	Open	Open	↓	24	↓	↓
P3	↓	4, 17, 24, 37	36	↓	26	↓	↓
P4	↓	4, 17, 24, 37	36	↓	26	↓	↓

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NOTES:
1. R1-R4 = 10 kΩ ±1%, 1/8 W.
2. For test conditions P3 and P4 adjust item GOI until each AMP output reads 6.9 to 7.1 Vrms at 1 kHz. Observe with item DOI that the output is an undistorted sine wave

* FIGURE 13. Power supply current test circuit for table X.

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TABLE XI. Small signal voltage gain check.

Test condition	Parameter	Item G01 frequency kHz ±1%	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
Q1	Small signal voltage gain check	0.10	6	5	0.95	1.05	V/V	14
Q2		1.0	6	5				
Q3		10.0	6	5				
Q4		100.0	6	5				
Q5		0.10	13	16				
Q6		1.0	13	16				
Q7		10.0	13	16				
Q8		100.0	13	16				
Q9		0.10	29	25				
Q10		1.0	29	25				
Q11		10.0	29	25				
Q12		100.0	29	25				
Q13		0.10	34	36				
Q14		1.0	34	36				
Q15		10.0	34	36				
Q16		100.0	34	36				

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TABLE XII. Large signal voltage gain check and output voltage swing test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
R1	Large signal voltage gain check	6	5	0.95	1.05	V/V	14
R2	↓	13	16	↓	↓	↓	↓
R3		29	25				
R4		34	36				
S1	Output voltage swing	6	5	7.0	15	Vpk	
S2	↓	13	16	↓	↓	↓	
S3		29	25				
S4		34	36				

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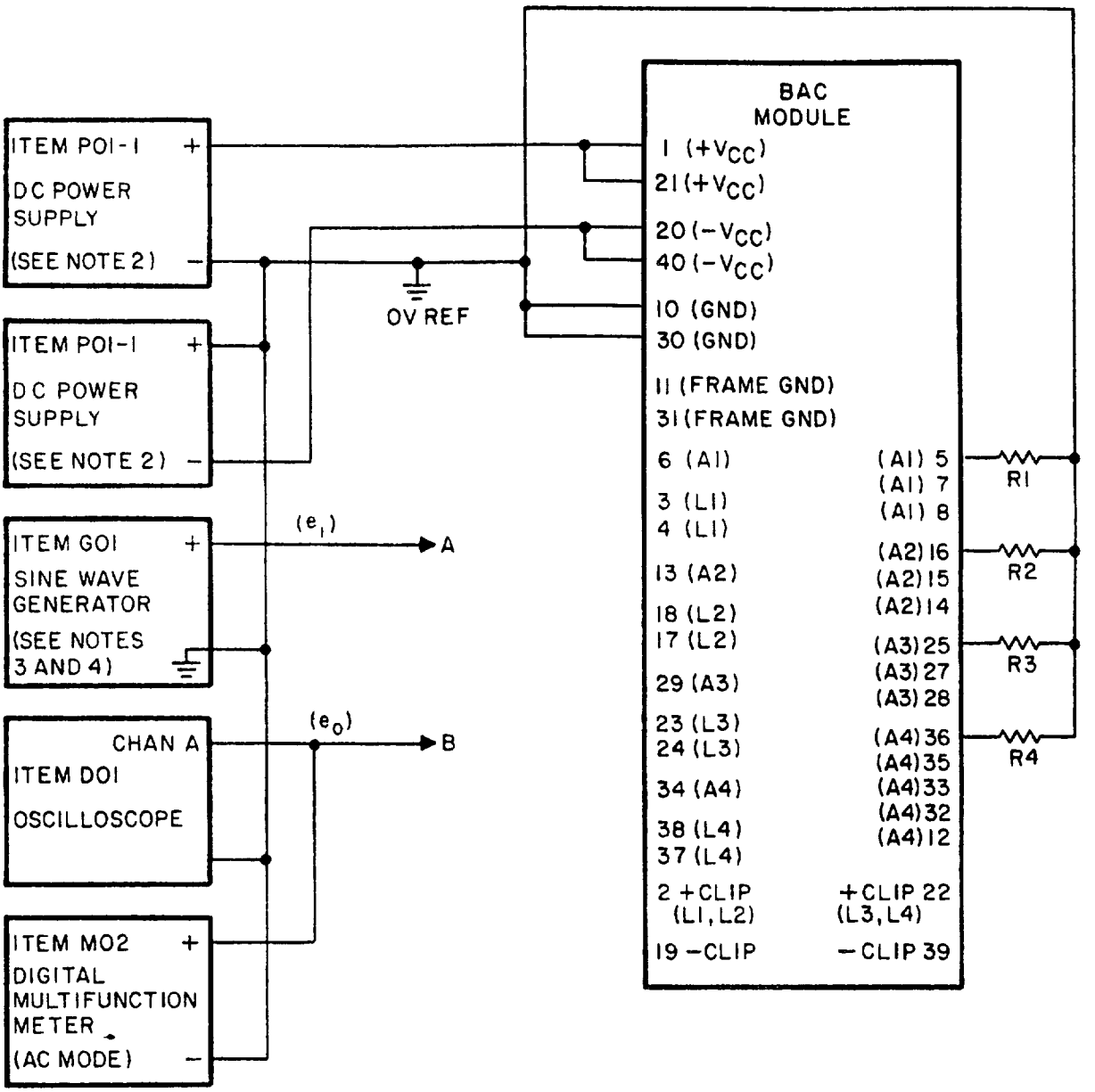


FIGURE 14. Small signal voltage gain check, large signal voltage gain check, and output voltage swing test circuit for tables XI and XII.

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NOTES:
1. R1-R4 = 10 kΩ ±1%, 1/8 W.
2. Set items P01-1 and P01-2 to 15 V ±1% for test conditions Q1-Q16 and R1-R4 and to 13.5 V ±1% for test conditions S1-S4.
3. Item G01 output parameters for test conditions Q1-Q16 and R1-R4:
   a. Test conditions Q1-Q16: e_i = 0.1000 ±0.001 Vrms.
   b. Test conditions R1-R4: e_i = 5.000 ±0.05 Vrms.
   c. Test conditions R1-R4: f = 1.0 kHz ±1%.
   d. Test conditions Q1-Q16 and R1-R4: Output waveform - sine wave.
4. For test conditions S1-S4, adjust item G01 until output sine wave (1 kHz ±1%) is clipped on both positive and negative peaks. Each peak shall ≥ 7.0 volts.
5. The output voltage limits for e_o is 0.95 Vrms minimum and 0.105 Vrms maximum for test conditions Q1-Q16. For test conditions S1-S4 the limits are 4.75 Vrms minimum and 5.25 Vrms maximum.
   Gain = e_o/e_i = 1.00 ±0.05 V/V.
   If e_o is within the specified limits, the gain will automatically be within limits.

FIGURE 14. Small signal voltage gain check, large signal voltage gain check, and output voltage swing test circuit for tables XI and XII - Continued.

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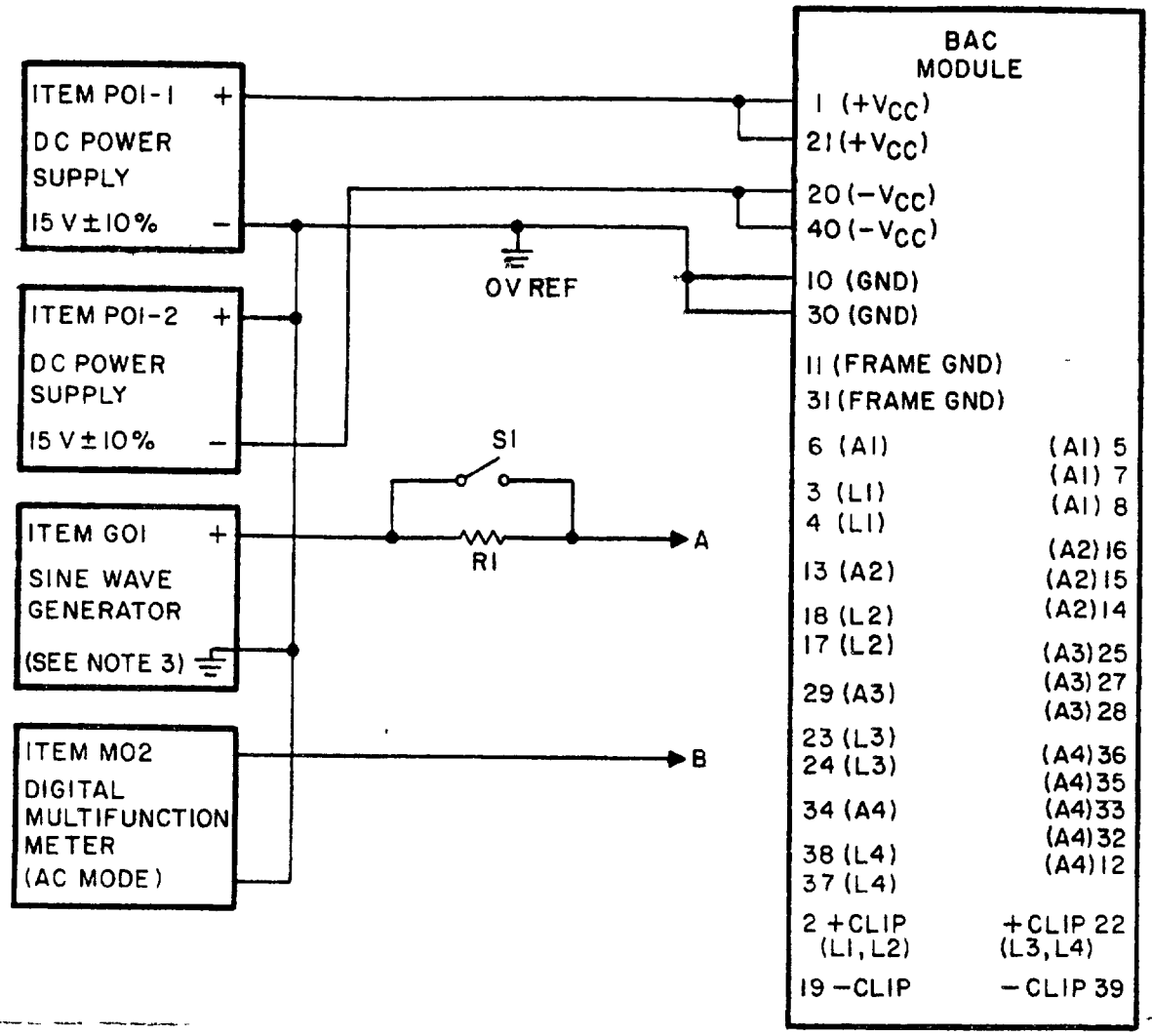
* TABLE XIII. Input resistance test.

Test condition	Parameter	Connect		Switch position	Output	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	S1		Min	Max		
V1	Input resistance	6	5	Closed	Set output to 1.00 ±0.01 Vrms				
		6	5	Open	Output to be >0.33 Vrms	5	∞	MΩ	15
V2		13	16	Closed	Set output to 1.00 ±0.01 Vrms				
		13	16	Open	Output to be >0.33 Vrms	5	∞	MΩ	15
V3		29	25	Closed	Set output to 1.00 ±0.01 Vrms				
		29	25	Open	Output to be >0.33 Vrms	5	∞	MΩ	15
V4		34	36	Closed	Set output to 1.00 ±0.01 Vrms				
		34	36	Open	Output to be >0.33 Vrms	5	∞	MΩ	15

TABLE XIV. Output noise test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
W1	Output noise	6	5	0.0	50	μVrms	16
W2		13	16				
W3		29	25				
W4		34	36				

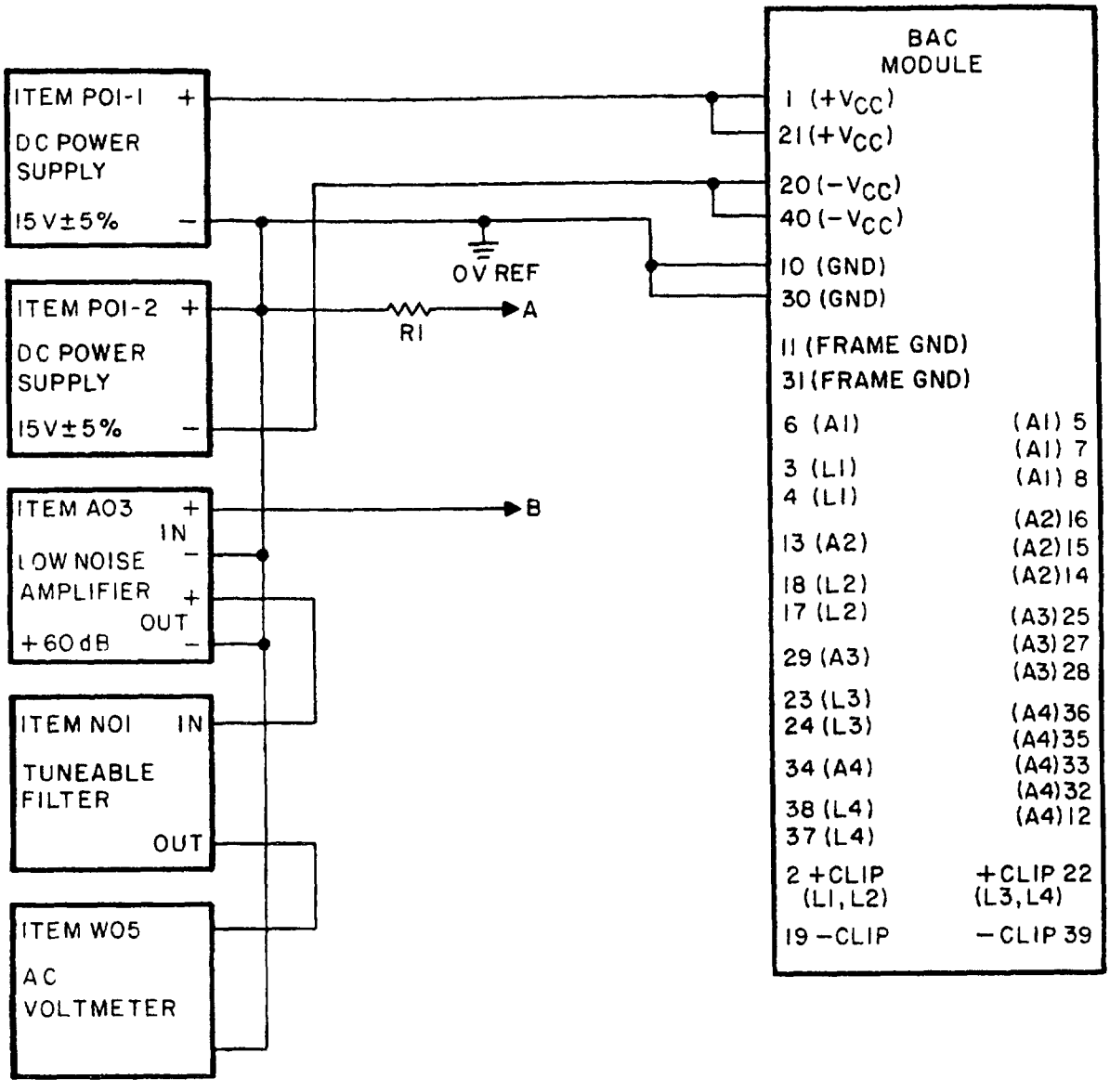
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NOTES:
1. R1 = 10 MΩ ±1%, 1/8 W.
2. S1 = SPST switch.
3. Set item G01 to 1.00 Vrms ±1% at 1 kHz.

*FIGURE 15. Input resistance test circuit for table XIII.

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NOTES:
1. R1 = 10 kΩ ±1%, 1/8 W.
2. Set item NO1 for band pass 100 Hz to 20 kHz.
3. Read true RMS AC with item WO5 divide by 1000.
4. Depending on conditions, the following steps may be required to reduce ambient or induced noise: (a) shield all lines and fixtures, and (b) substitute batteries for items PO1-1 and PO1-2.

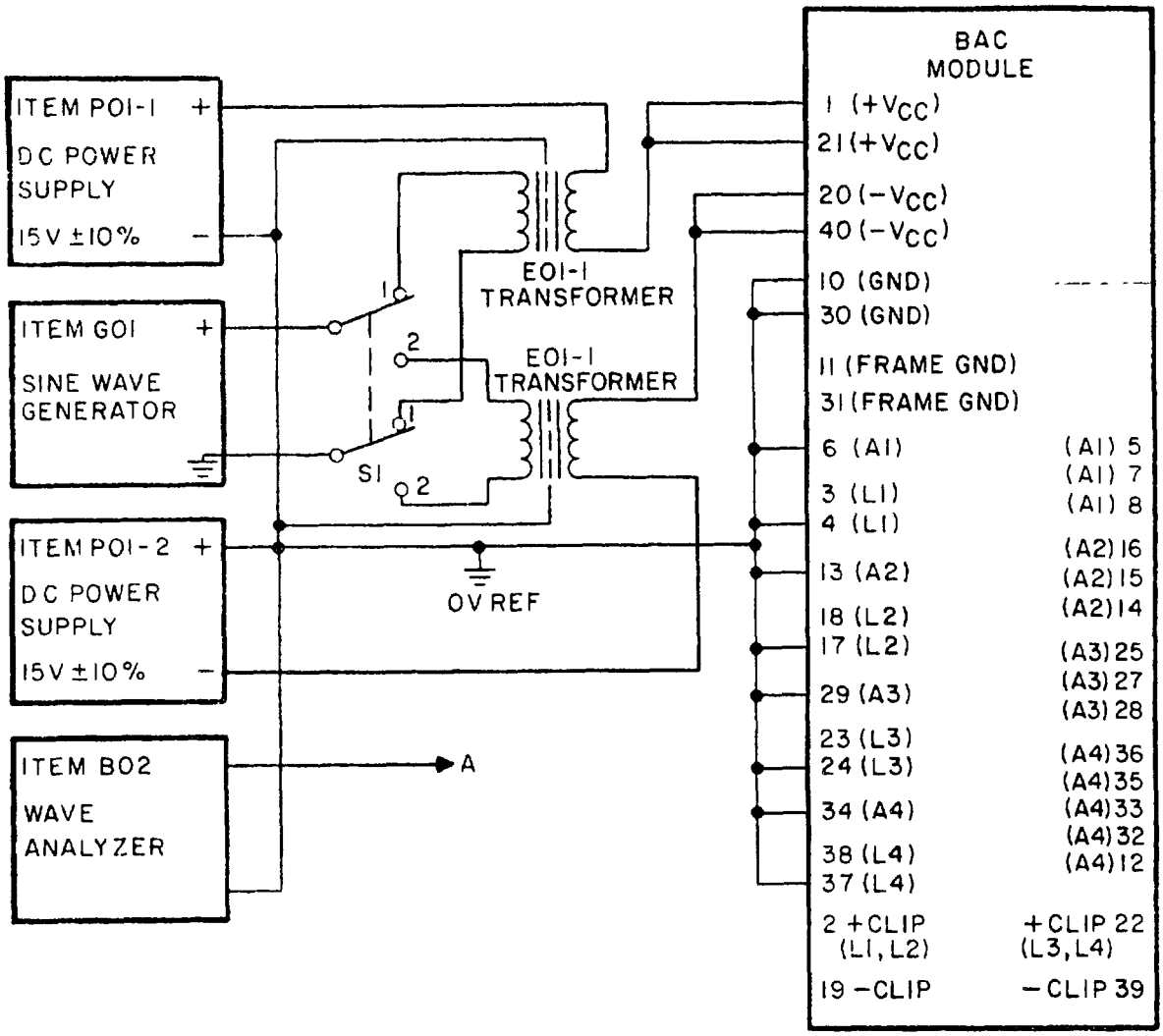
FIGURE 16. Output noise test circuit for table XIV.

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TABLE XV. Power supply rejection ratio check.

Test condition	Parameter	Connect A to pin	Initial limits 25°+5°C ambient Min	Initial limits 25°+5°C ambient Max	Unit	Test figure
X1	Power supply rejection ratio check	5	60	∞	dB	17
X2		16				
X3		25				
X4		36				
X5		5				
X6		16				
X7		25				
X8		36				

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NOTES:
1. S1 = DPDT switch.
2. Set item G01 to 1 Vrms ±1% at 800 Hz ±10%
3. Power supply rejection ratio = $20 \log \frac{e_{AC}}{e_{OUT}} = 20 \log \frac{1.0\ V}{e_{OUT}}$

   where e_{OUT} = 1.0 mVrms maximum.

FIGURE 17. Power supply rejection ratio check test circuit for table XV.

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TABLE XVI. Functional without and with input limiting test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Y1	Functional without input limiting	4	7	2.40	2.60	Vrms	18
Y2	↓	17	15	↓	↓	↓	↓
Y3	↓	24	27	↓	↓	↓	↓
Y4	↓	37	35	↓	↓	↓	↓
Y5	Functional with input limiting	4	7	4.9	6.3	Vpk	↓
Y6	↓	17	15	↓	↓	↓	↓
Y7	↓	24	27	↓	↓	↓	↓
Y8	↓	37	35	↓	↓	↓	↓

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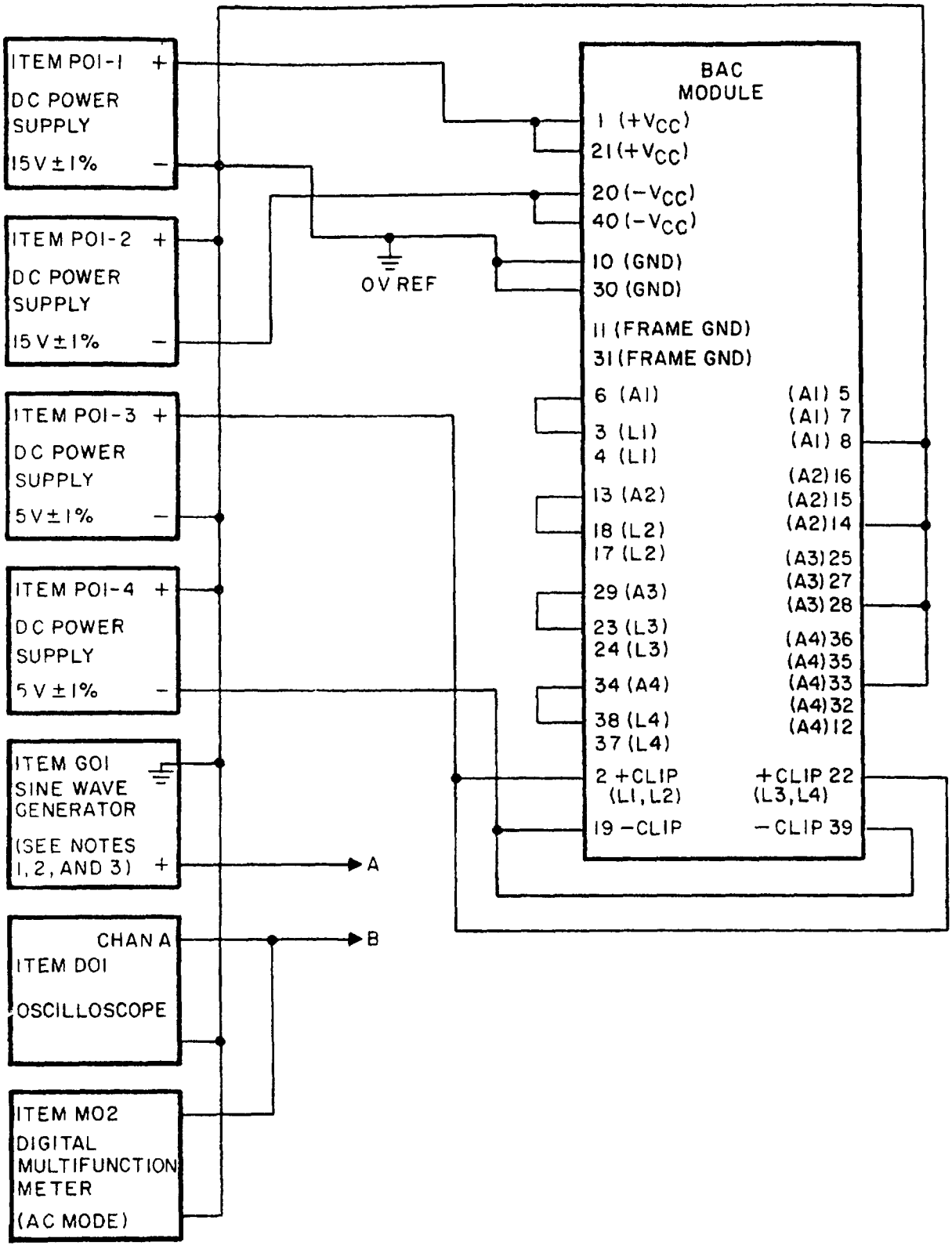


*FIGURE 18. Functional without and with input limiting test circuit for table XVI.

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NOTES:
1. Set item G01 to 1000 Hz.
2. Set item G01 to 2.5 Vrms ±1% for test conditions Y1-Y4 (output waveform - sine wave).
3. Set item G01 to 10.0 Vrms ±10% for test conditions Y5-Y8 (output waveform - symmetrically clipped sine wave).
4. For test conditions Y5-Y8 + E_p and - E_p shall meet the requirements of table XVI.

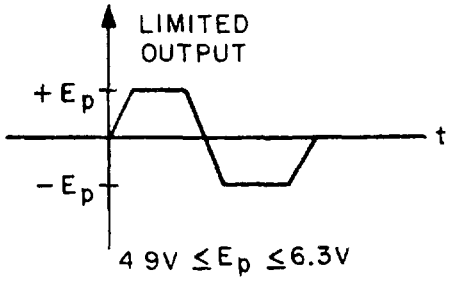


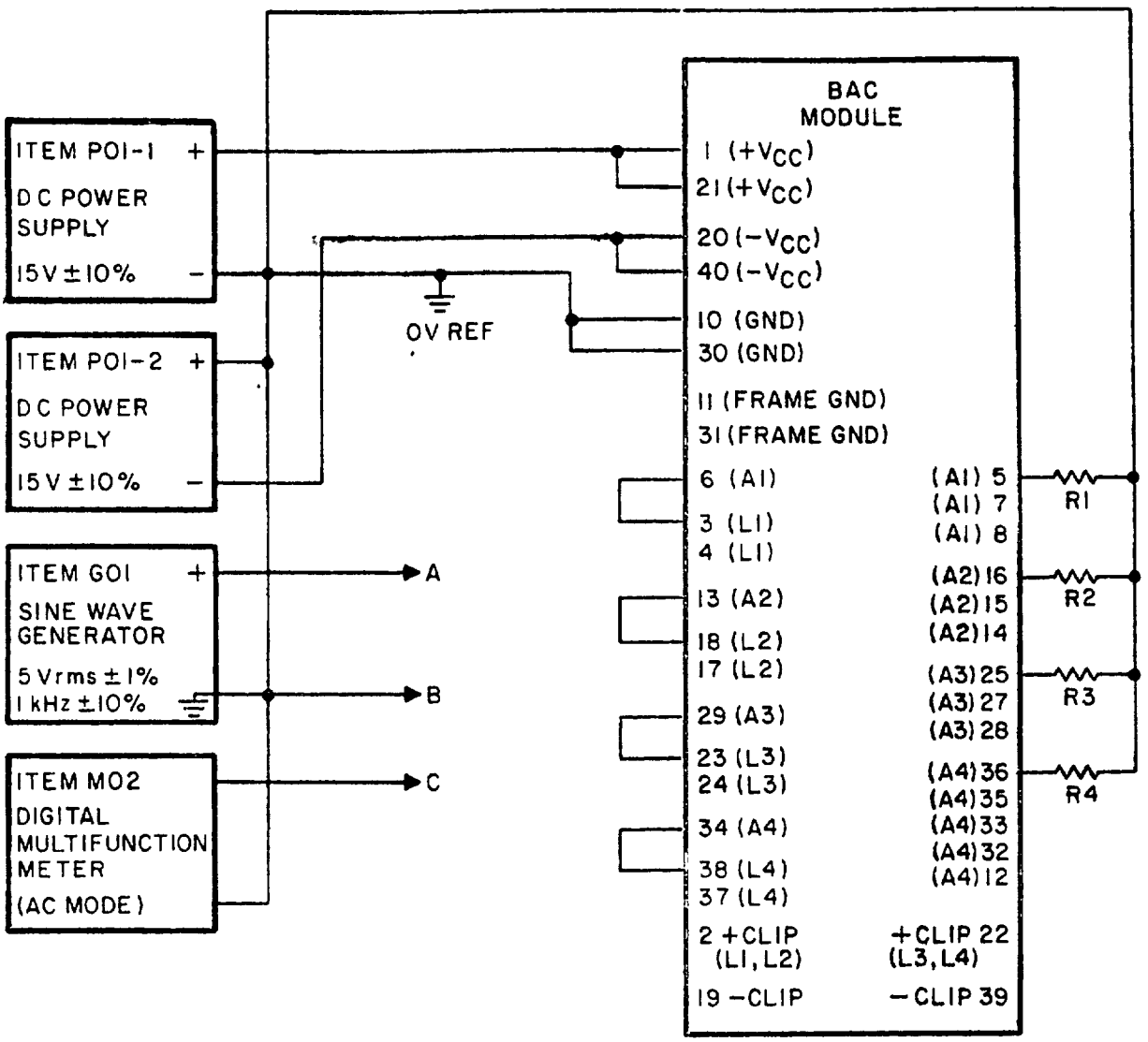
FIGURE 18. Functional without and with input limiting test circuit for table XVI - Continued.

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TABLE XVII. Crosstalk test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Z1	A1 to A2 Crosstalk	4	17, 24, 37	16	60	∞	dB	19
Z2	A1 to A3	4	17, 24, 37	25				
Z3	A1 to A4	4	17, 24, 37	36				
Z4	A2 to A1	17	4, 24, 37	5				
Z5	A2 to A3	17	4, 24, 37	25				
Z6	A2 to A4	17	4, 24, 37	36				
Z7	A3 to A1	24	4, 17, 37	5				
Z8	A3 to A2	24	4, 17, 37	16				
Z9	A3 to A4	24	4, 17, 37	36				
Z10	A4 to A1	37	4, 17, 24	5				
Z11	A4 to A2	37	4, 17, 24	16				
Z12	A4 to A3	37	4, 17, 24	25				

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NOTES:
1. R1-R4 = 10 kΩ ±1%, 1/8 W.
2. Maximum output = 5.0 mVrms.

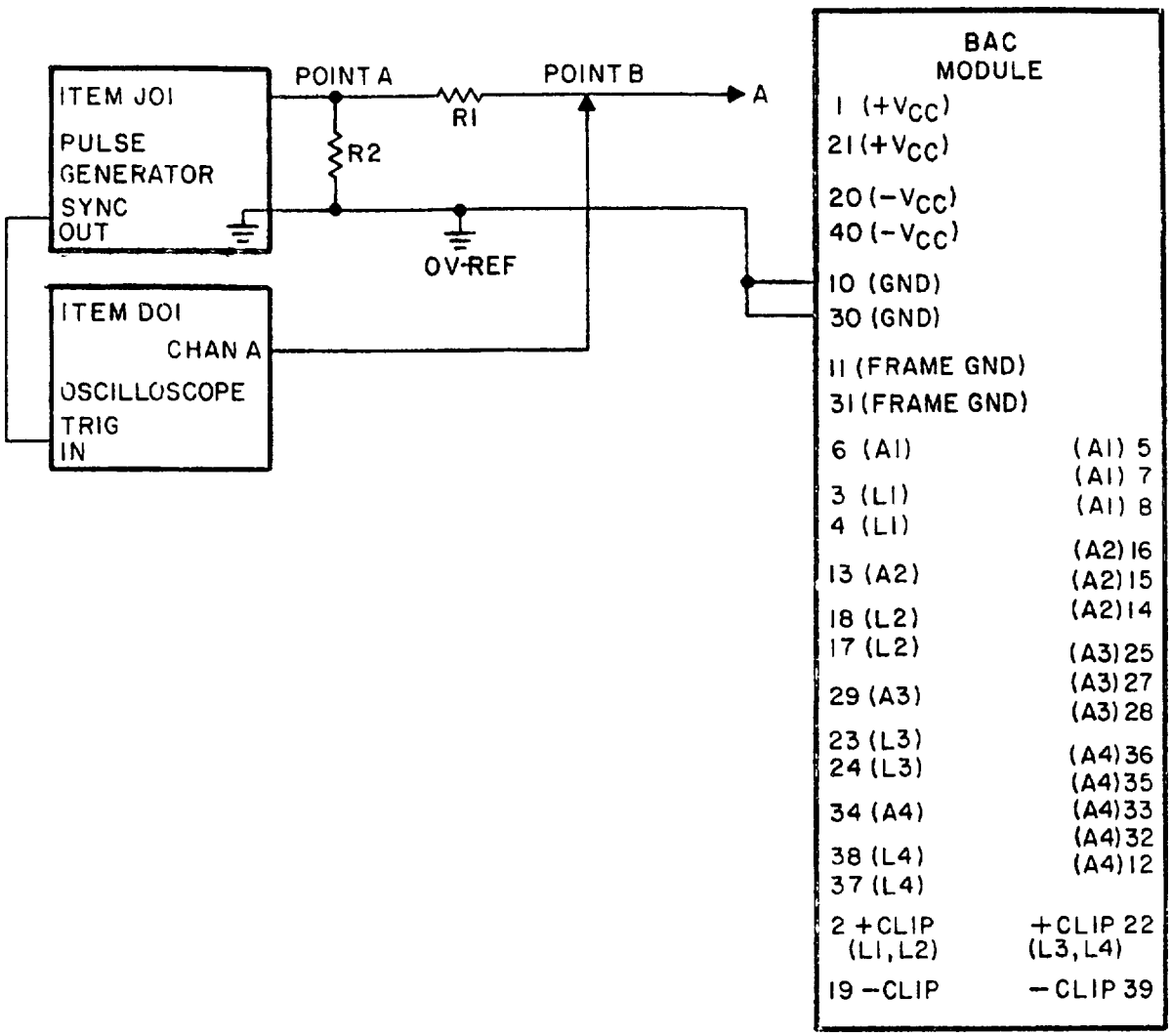
*FIGURE 19. Crosstalk test circuit for table XVII.

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TABLE XVIII. Power supply transient rise time test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
AA1	Power supply transient rise time	1	31	93	µs	20, 21
AA2	Power supply transient rise time	20	31	93	µs	20, 21

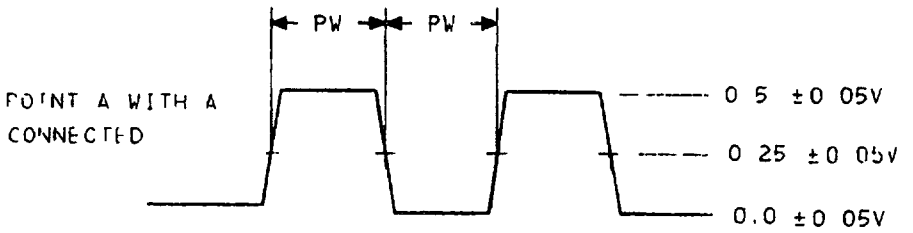
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NOTES:
1. R1 = 412Ω ±1%, 1/8 W.
2. R2 = 50Ω ±5%, 1 W.

FIGURE 20. Power supply transient rise time test circuit for table XVIII.

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NOTE Input pulse parameters
PW = 300 ±3 µs
$t_{TLH} = t_{THL}$ = 0 50 ±0.05 µs.

FIGURE 21 Power supply transient rise time test circuit waveforms for table XVIII

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